

ACHAL MUKKAPATI

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EDUCATION

Northeastern University

Master of Science in Artificial Intelligence, Concentration in Machine Learning

Boston, Massachusetts

GPA: 4.00 05/2027

Rutgers University

Bachelor of Science in Computer Science and Business Analytics & Information Technology (BAIT)

New Brunswick, New Jersey

05/2025

TECHNICAL SKILLS

Languages/Frameworks: Python, Java, SQL, C/C++, TypeScript, Bash, Korn, HTML/CSS, JavaScript, R, Flask

Libraries: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV, SciPy

Technologies/Tools: Git, Jenkins, Docker, Kubernetes, Linux/Unix, MERN Stack, MySQL, Jupyter Notebook

Coursework/Certifications: AWS Certified Cloud Practitioner, Data Structures, Database Design, Algorithms, Deep Neural Networks, Natural Language Processing, Computer Architecture, Operating Systems, Agile Methodology

EXPERIENCE

General Motors - Cruise

AI/ML Software Engineering Intern

Sunnyvale, California

05/2026 – 08/2026

- Architected a **multimodal vision-language model (VLM)**-based validation pipeline for autonomous vehicle simulation, converting road and simulation video into side-by-side artifacts, to improve scenario review and analysis
- Implemented a **simulation fidelity benchmarking framework** for autonomous perception systems, enabling scalable validation and root-cause analysis across vehicle behavior, road geometry, traffic controls, and visual defects
- Accelerated review of 2,000 AV scenarios, surfacing 100 priority simulations and cutting triage effort by 95%

People Tech Group

Software Engineering Intern

Redmond, Washington

06/2024 – 08/2024

- Automated **Infrastructure as Code (IaC)** pipelines to convert architecture diagrams into **CloudFormation** stacks
- Invoked **Amazon Bedrock LLM agents** to analyze architecture diagrams from **S3 buckets** and generate configuration prompts, using **Lambda functions** to process input and produce tailored IaC
- Delivered **SDLC** improvements through GitHub-based IaC collaboration, streamlining peer reviews, orchestrating testing with **CodePipeline**, cutting provisioning time by 75% and accelerating **CI/CD** deployment cycles

KBC Bank NY

Information Technology Intern

New York, New York

05/2023 – 08/2023

- Tested and tuned **SQL queries** for report generation and built **Linux shell scripts** to email Oracle alert log errors, monitor space usage, and track backup status using **Cron scheduling**, improving database infrastructure efficiency
- Designed **Bash** and **Korn** workflows to eliminate repetitive tasks, saving 2.5 hours daily and minimizing human error across infrastructure monitoring, database maintenance, and system health checks in a production environment
- Shadowed the senior architect, applying best practices in **big data processing** to enterprise financial systems

PROJECTS

Context-Switching RPG Agent - Python, Lua, Perceptrons, Hebbian Learning, Markov Models, A* Search

- Developed an interpretable **autonomous game-playing agent** for Pokémon FireRed using context-specific perceptron pipelines, curiosity-driven exploration, and Hebbian adaptation to support long-horizon decision making
- Engineered a **checkpoint-aware demonstration retrieval** framework leveraging Markov state transitions and progress-scoped matching, achieving autonomous progression across 11 game maps and a 77.8% battle win rate

Metadata Filtering for Doctor-Patient Access Control - Python, Amazon Bedrock, DynamoDB, Lambda

- Established a metadata-filtered retrieval layer for **Amazon Bedrock Knowledge Bases** using DynamoDB and AWS Lambda, enforcing doctor-patient authorization policies and reducing unauthorized data access incidents by 30%
- Enabled secure **natural-language retrieval of healthcare records** through an interactive AI assistant built with Amazon Q Developer, improving control-check efficiency by 33%